Tarea I Métodos Numericos II C00368

Problema 1: $y'' = \frac{2y}{x^2} - \frac{1}{x}$ con $y(x_0)=\alpha \wedge y(x_N)=\beta$

Donde $p(x)=0$, $q(x)=\frac{2}{x^2}$, $r(x)=\frac{-1}{x}$, además $h=\frac{x_f-x_i}{N+1}$

a) La matriz general tiene la forma:

Con
- la diagonal principal $A_{i,i} = 2+h^2 q(x_{i+1}) = 2+\frac{2h^2}{x^2} = 2\left(1+\frac{h^2}{x^2}\right)$
- la diagonal superior $A_{i,i+1} = -1+\frac{h}{2}p(x_{i+1})$
- la diagonal inferior $A_{i+1,i} = -1-\frac{h}{2}p(x_{i+1})$

} con $p(x)=0$, ambas diagonales superior e inferior son -1

$$A = \begin{bmatrix} 2\left(1+\frac{h^2}{x_2^2}\right) & -1 & 0 & \cdots & 0 \\ -1 & 2\left(1+\frac{h^2}{x_3^2}\right) & -1 & & \vdots \\ 0 & -1 & \ddots & \ddots & -1 \\ \vdots & \vdots & & & \\ 0 & 0 & \cdots & -1 & 2\left(1+\frac{h^2}{x_{N+1}^2}\right) \end{bmatrix}_{N\times N}$$

y el vector $\vec{b}$ tiene la forma: con $p(x)=0 \wedge r=\frac{-1}{x}$

Para
- $i=1$: $b_1 = \left[1+\frac{h}{2}\cancel{p(x_2)}^{\,0}\right]\alpha - h^2\cdot r(x_2) = \frac{h^2}{x_2}+\alpha$
- $i=i$: $b_i = -h^2\cdot r(x_{i+1}) = \frac{h^2}{x_{i+1}}$
- $i=N$: $b_N = \left[1-\frac{h}{2}\cancel{p(x_{N+1})}^{\,0}\right]\beta - h^2\cdot r(x_{N+1}) = \frac{h^2}{x_{N+1}}+\beta$

$$\vec{b} = \begin{bmatrix} \frac{h^2}{x_2}+\alpha \\ \vdots \\ \frac{h^2}{x_{i+1}} \\ \vdots \\ \frac{h^2}{x_{N+1}}+\beta \end{bmatrix}_{N\times 1}$$

b) El sistema queda: $Ay = b$,

$$\begin{bmatrix} 2\left(1+\frac{h^2}{x_2^2}\right) & -1 & 0 & \cdots & 0 \\ -1 & 2\left(1+\frac{h^2}{x_3^2}\right) & -1 & & \vdots \\ 0 & -1 & \ddots & \ddots & -1 \\ \vdots & \vdots & & & \\ 0 & 0 & \cdots & -1 & 2\left(1+\frac{h^2}{x_{N+1}^2}\right) \end{bmatrix}_{N\times N} \begin{bmatrix} y_1 \\ \vdots \\ \vdots \\ y_N \end{bmatrix} = \begin{bmatrix} \frac{h^2}{x_2}+\alpha \\ \vdots \\ \frac{h}{x_{i+1}} \\ \vdots \\ \frac{h^2}{x_{N+1}}+\beta \end{bmatrix}_{N\times 1}$$

c) Ver código adjunto

Problema 2: Sea $y'' = -2\sec(x)y'^3 - y^2\sec(x)y' = F(x,y,y')$

Y sus derivadas Respecto a y: $-2\sec(x)yy'$ $\wedge$ respecto a y': $-6\sec(x)y'^2 - \sec(x)y^2$

$y(x_0)=\alpha \quad \wedge \quad y(x_N)=\beta$

a)

Para
$$\begin{cases} i=1: & y' = \frac{y_2-\alpha}{2h} \implies -\alpha + 2y_1 - y_2 + h^2 F\left(x_1, y_1, \frac{y_2-\alpha}{2h}\right) = 0 \\ i=i: & y' = \frac{y_{i+1}-y_{i-1}}{2h} \implies -y_{i-1} + 2y_i - y_{i+1} + h^2 F\left(x_i, y_i, \frac{y_{i+1}-y_{i-1}}{2h}\right) = 0 \\ i=N: & y' = \frac{\beta - y_{N-1}}{2h} \implies -\beta - y_{N-1} + 2y_N + h^2 F\left(x_N, y_N, \frac{\beta - y_{N-1}}{2h}\right) = 0 \end{cases}$$

b)

Para la diagonal principal de J tenemos

$J_{(i,i)} = 2 + h^2 \cdot F_y(x_{i+1}, y_{i+1}, y') = 2 - 2h^2\sec(x_{i+1})y_{i+1}y'$, con y' definido anteriormente

Para la diagonal superior e inferior

$$J_{(i,i+1)} = -1 + \frac{h}{2}F_{y'}(x_{i+1}, y_{i+1}, y') = -1 - \frac{h\sec(x_{i+1})}{2}\left[6y'^2 + y_{i+1}^2\right]$$

$$J_{(i+1,i)} = -1 - \frac{h}{2}F_{y'}(x_{i+1}, y_{i+1}, y') = -1 + \frac{h\sec(x_{i+1})}{2}\left[6y'^2 + y_{i+1}^2\right]$$

Con y' definido anteriormente para $i=1$, $i=i$ y $i=N$

La Matriz J queda como:

$$J = \begin{bmatrix} 2-2h^2\sec(x_{i+1})y_{i+1}\left(\frac{y_2-\alpha}{2h}\right) & -1-\frac{h\sec(x_{i+1})}{2}\left[6\left(\frac{y_2-\alpha}{2h}\right)^2+y_{i+1}^2\right] & \cdots & & 0 \\ -1+\frac{h\sec(x_{i+1})}{2}\left[6\left(\frac{y_2-\alpha}{2h}\right)^2+y_{i+1}^2\right] & \ddots & \ddots & & \vdots \\ \vdots & & 2-2h^2\sec(x_{i+1})y_{i+1}\left(\frac{y_{i+1}-y_{i-1}}{2h}\right) & -1-\frac{h\sec(x_{i+1})}{2}\left[6\left(\frac{y_{i+1}-y_{i-1}}{2h}\right)^2+y_{i+1}^2\right] & \\ & & -1+\frac{h\sec(x_{i+1})}{2}\left[6\left(\frac{y_{i+1}-y_{i-1}}{2h}\right)^2+y_{i+1}^2\right] & \ddots & \vdots \\ & & & \ddots & -1-\frac{h\sec(x_{i+1})}{2}\left[6\left(\frac{\beta-y_{N-1}}{2h}\right)^2+y_{i+1}^2\right] \\ 0 & \cdots & & -1+\frac{h\sec(x_{i+1})}{2}\left[6\left(\frac{\beta-y_{N-1}}{2h}\right)^2+y_{i+1}^2\right] & 2-2h^2\sec(x_{i+1})y_{i+1}\left(\frac{\beta-y_{N-1}}{2h}\right) \end{bmatrix}_{N\times N}$$

El vector F se forma talque:

$$\vec{F} = \begin{bmatrix} -\alpha+2y_1-y_2-h^2\sec(x_1)\left[2\left(\frac{y_2-\alpha}{2h}\right)^3+y_1^2\left(\frac{y_2-\alpha}{2h}\right)\right] \\ \vdots \\ -y_{i-1}+2y_i-y_{i+1}+h^2\sec(x_i)\left[2\left(\frac{y_{i+1}-y_{i-1}}{2h}\right)^3+y_i^2\left(\frac{y_{i+1}-y_{i-1}}{2h}\right)\right] \\ \vdots \\ -y_{N-1}+2y_N-\beta+h^2\sec(x_N)\left[2\left(\frac{\beta-y_{N-1}}{2h}\right)^3+y_N^2\left(\frac{\beta-y_{N-1}}{2h}\right)\right] \end{bmatrix}_{N\times 1}$$

y con ello se forma el sistema $Jv=-F$

c) Ver codigo adjunto